Tactix interim audit: online configuration boundary

Codex for Andrew

25 September 2026

Outcome. The codebase is not yet ready for production online learning. This source-only checkpoint changed no solver or acceptance contract. The detailed evidence is in the AM438 technical report under `benchmarks/reports/`.

Configuration lifetime. All three public online-configuration entry points finalize C input before rejecting an invalid program. The online executor's own replacement is transactional, but the surrounding input session can become closed: later checked clause additions then return -1. The C header does not promise rollback on rejection, so this is a contract risk, not a claimed regression. A simple call reorder is unsafe because input finalization retires absent variables and settles proof identities used by program conversion. A design should specify admission versus finalization and cover valid and invalid v1–v3 programs with attached and unattached proofs.

Learning architecture. A synchronous solve currently creates a fresh stack-local DefaultPolicy. The persistent C session owns a separate online inprocessing executor, not the running default policy. For in-run adjustment of pass dose/order and switching branching or clause-deletion substrategies, the next structural work is a session-owned policy with default-exact safe-point slots and owned strategy arms. The current reduction seam chooses victims only; collection and tier decisions remain embedded in DefaultPolicy. Stage this behind a byte-identical default trace gate, then check proofs, models, public reachability, fuzz, and matched performance at each cutover. This is a recommendation, not an implemented improvement.

Correctness gates. Current-source Debug job 1723391 had passed 186 of 187 tests at this snapshot and was still running the final scripted demo. Both AM434 extended AES proof-checking tasks were still running. Their terminal results must be inspected; no acceptance follows from partial passes. The reduced H2 native DSR still fails the canonical no-option checker, so N2 remains open. N1, N3, N4, affected fuzz, and final learning readiness also remain open. The requested 1,166-instance Tier D, 5,000-second, 4-CPU/32-GiB hwulab comparison is correctly deferred until those goals are actually verified.

Next actions. Finish the running gates; resolve the general H2 proof/checker-state discrepancy under a chosen canonical contract; then land the staged policy/session refactor with exact default-parity checks. Do not promote a partial checker result or run Tier D early.